DO NOW · AO1 REVISION

Background radiation, different half-lives and uses of nuclear radiation

Atomic structure
Paper 1 · Separate Physics only
12 questions

Cover the answers. Answer each question out loud or on paper, then check. Tick the ones you know. Learn the 3 starred (★) questions first.

	QUESTION	ANSWER	✓
1 ★	**What is background radiation?**	Radiation that is around us all the time, from natural and human-made sources.	☐
2	**Name two natural sources of background radiation.**	Rocks and cosmic rays	☐
3 ★	**Name two human-made sources of background radiation.**	Fallout from nuclear weapons testing and nuclear accidents	☐
4	**What is radiation dose?**	A measure of the risk of harm from exposure to radiation.	☐
5	**Explain why an isotope with a short half-life can be very hazardous at first.**	Its nuclei decay quickly, so its activity is very high.	☐
6	**Give two uses of nuclear radiation in medicine.**	Exploring internal organs, and controlling or destroying unwanted tissue	☐
7	**Explain why alpha radiation is not suitable for exploring internal organs.**	It cannot pass out of the body to be detected, and it is strongly ionising.	☐
8	**State the unit of radiation dose.**	Sievert (Sv)	☐
9	**State how many millisieverts (mSv) equal 1 sievert (Sv).**	1000 mSv	☐
10	**Give two things that affect a person's background radiation dose.**	Their occupation and their location	☐
11 ★	**Explain why a medical tracer should have a short half-life.**	Its activity falls quickly, limiting the dose to the patient.	☐
12	**Explain why a medical tracer should emit gamma radiation.**	It passes out of the body to be detected and is only weakly ionising.	☐

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